

Mohammed Aharrar Soulali

Machine Learning Engineer

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PROFILE

Recent Master’s-level engineering graduate from **École des Mines de Saint-Étienne**, specialized in **Data Science** and **Computer Science**. Projects in NLP, computer vision, and AWS, with a foundation in software engineering. **Available immediately**.

EXPERIENCE

Final-Year Internship – Machine Learning Applied to Technical Reports <i>EDF</i>	May 2025 – September 2025 <i>Paris, France</i>
- Designed an NLP POC on 17,000 French technical reports to predict importance levels and assist report writing - Evaluated TF-IDF, Word2Vec, and transformer approaches for classification and similar-report suggestion	
Assistant Engineer Internship <i>École des Mines de Saint-Étienne Research Center</i>	June 2024 – August 2024 <i>Saint-Étienne, France</i>
- Developed a Python computer vision system for quality control on an automated pot-filling line - Implemented Canny algorithms, ellipse detection, feature extraction, and optimization to estimate pot geometry	

EDUCATION

Engineering Degree – Civil Engineer of Mines <i>École des Mines de Saint-Étienne</i>	September 2022 – September 2025 <i>Saint-Étienne, France</i>
Data Science: statistical learning machine learning time series optimization Computer Science: software engineering security and privacy cloud & edge infrastructures AI / Big Data: knowledge representation simulation information systems data visualization - Minors: Image and Pattern Recognition High-Performance Programming Project Management	
Bachelor’s Degree in Mathematics <i>Jean Monnet University</i>	September 2023 – June 2024 <i>Saint-Étienne, France</i>
Mathematics degree completed in parallel with the engineering degree, strengthening foundations in analysis, algebra, probability and statistics.	

PROJECTS

Industrial Anomaly Detection <i>Python, R, FPCA</i>	February 2025 – April 2025
- Analyzed production sound signals to detect anomalies in an automated filling process - Benchmarked PCA, FPCA, OSPCA, and Mahalanobis methods; best F1-score: 0.92	
Industrial Video Analysis with Orano <i>Python, Computer Vision</i>	September 2024 – January 2025
- Automated inspection-video analysis to quantify powder build-up in pneumatic transport tubes - Used image filtering and geometric indicators to locate affected areas and provide a quality diagnosis	
IoT Cloud Pipeline <i>AWS, Java, S3, SQS, Lambda</i>	November 2024 – January 2025
- Designed an AWS pipeline to ingest, process, and consolidate IoT data - Compared server-based and serverless architectures using S3, SQS, EC2, Lambda, and CloudWatch	
Multi-Agent Traffic Optimization <i>Java, Simulation, AI</i>	January 2024 – June 2024
- Built a Java simulation of an intersection with intelligent traffic lights to reduce vehicle waiting time - Implemented decentralized decisions based on cumulative waiting time; reduced delays by a factor of 2 to 4 depending on scenarios	

LANGUAGES

French: B2	English: C1 (TOEIC 945, 2024)	Arabic: C1 (Native language)	German: A1+
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SKILLS

Programming: Python, Java, SQL, PostgreSQL, R, Bash/Shell
ML / AI: scikit-learn, TensorFlow, NLP, Transformers, RAG, LangChain, FAISS, Ollama, LSTM, Computer Vision
Cloud / DevOps / Backend: AWS, Flask, FastAPI, REST APIs, Git/GitHub, Docker, CI/CD
Other: Scrum, Problem-solving, Teamwork, Adaptability